1. Algebra

1.1. The Modulus Function

- It gives the absolute value of a number.
- The modulus of a value gives the distance of the value from the origin.
- No line with a modulus ever goes under the x-axis.
- Any line that does go below the x-axis, when modulated is reflected above it.

$$|a \times b| = |a| \times |b|$$

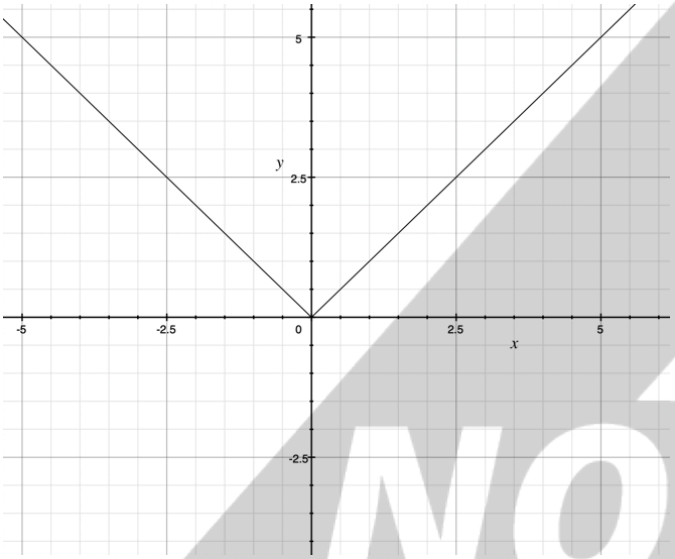
$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$$

$$|x^2| = |x|^2 = x^2$$

$$|x| = |a| \Leftrightarrow x^2 = a^2$$

$$\sqrt{x^2} = |x|$$

- Graph of y= $|x|$



1.2. Polynomials

- To find unknowns in a given identity, either
  - Substitute suitable values of x
  - Equalize given coefficients of like powers of x
- **Factor theorem:** If $(x-t)$ is a factor of the function $p(x)$ then $p(t)=0$
- **Remainder theorem:** If the function $f(x)$ is divided by $(x-t)$ then the remainder: $R=f(t)$

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

1.3. Binomial Series

Expanding $(1+x)^n$ where $\|x\| < 1$

$$1 + \frac{n}{1}x + \frac{n(n-1)}{1 \times 2}x^2 + \frac{n(n-1)(n-2)}{1 \times 2 \times 3}x^3 + \ldots$$

- **Factor case:** if constant is not 1, pull out a factor from brackets to make it 1 & use general equation. Do not forget the indices.
- **Substitution case:** if bracket contains more than one x term (e.g. $(2-x+x^2)$) then make the last part u, expand and then substitute back in.
- **Finding the limit of x **in expansion:**
  **E.g. $(1+ax)^n$, limit can be found by substituting ax between the modulus sign in $\|x\| < 1$ and altering it to have only x in the modulus

{S15-P31} Question 3:

Show that, for small values of x^2,

$$(1-2x^2)^{-2} - (1+6x^2)^{\frac{2}{3}} \approx kx^4,$$

where the value of the constant k is to be determined.

Solution:

Expand $(1-2x^2)^{-2}$ until the x^4 term

$$(1+x)^{-2} = 1 + (-2)x + \frac{-2((-2)-1)}{1 \times 2}x^2$$

$$= 1 - 2x + 3x^2$$

$$(1-2x^2)^{-2} = 1 - 2(2x^2) + 3(2x^2)^2 = 1 + 4x^2 + 12x^4$$

Expand $(1+6x^2)^{\frac{2}{3}}$ until the x^4 term

$$(1+x)^{\frac{2}{3}} = 1 + \left(\frac{2}{3}\right)x + \frac{\frac{2}{3}\left(\left(\frac{2}{3}\right)-1\right)}{1 \times 2}x^2$$

$$= 1 + \frac{2}{3}x - \frac{1}{9}x^2$$

$$(1+6x^2)^{\frac{2}{3}} = 1 + \frac{2}{3}(6x^2) - \frac{1}{9}(6x^2)^2 = 1 + 4x^2 - 4x^4$$

Subtract the terms of the expansion of $(1+6x^2)^{\frac{2}{3}}$ from those of $(1-2x^2)^{-2}$

$$(1 - 4x^2 + 12x^4) - (1 + 4x^2 - 4x^4)$$

$$= -8x^2 + 16x^4$$

The value of k is:

$$16$$

1.4. Partial Fractions

$$\frac{ax+b}{(px+q)(rx+s)} \equiv \frac{A}{px+q} + \frac{B}{rx+s}$$

- Multiply $(px+q)$, substitute $x = -\frac{q}{p}$ and find A
- Multiply $(rx+s)$, substitute $x = -\frac{s}{r}$ and find B

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$$\frac{ax^2+bx+c}{(px+q)(rx+s)^2} \equiv \frac{A}{px+q} + \frac{B}{rx+s} + \frac{C}{(rx+s)^2}$$

- Multiply $(px+q)$, substitute $x=-\frac{q}{p}$ and find A
- Multiply $(rx+s)^2$, substitute $x=-\frac{s}{r}$ and find C
- Substitute any constant e.g. $x=0$ and find B

$$\frac{ax^2+bx+c}{(px+q)(rx^2+s)} \equiv \frac{A}{px+q} + \frac{Bx+C}{rx^2+s}$$

- Multiply $(px+q)$, substitute $x=-\frac{q}{p}$ and find A
- Take $\frac{A}{px+q}$ to the other side, subtract and simplify.
- Linear eqn. left at top is equal to $Bx+C$
- **Improper fraction case:** if numerator has x to the degree of power equivalent or greater than the denominator then another constant is present. This can be found by dividing denominator by numerator and using remainder

{S12-P33} Question 8:

Express the following in partial fractions:

$$\frac{4x^2-7x-1}{(x+1)(2x-3)}$$

Solution:

Expand the brackets

$$\frac{4x^2-7x-1}{2x^2-x-3}$$

Greatest power of x same in numerator and denominator, thus is an improper fraction case

Making into proper fraction:

$$\begin{array}{r|l} & 2 \\ \hline 2x^2-x-3 & 4x^2-7x-1 \\ & \underline{4x^2-2x-6} \\ & -5x+5 \end{array}$$

This is written as:

$$2+\frac{5-5x}{(x+1)(2x-3)}$$

Now proceed with normal case for the fraction:

$$\frac{A}{x+1}+\frac{B}{2x-3}=\frac{5-5x}{(x+1)(2x-3)}$$

$$A(2x-3)+B(x+1)=5-5x$$

When $x=-1$

$$5A=5+5$$

$$A=-2$$

When $x=\frac{3}{2}$

$$\frac{5}{2}B=5-\frac{15}{2}$$

$$B=-1$$

Thus, the partial fraction is:

$$2+\frac{-2}{x+1}+\frac{-1}{2x-3}$$

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